Niccolò Zuppichini

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EXPERIENCE

Lemonate April 2025 – Present
Senior Software Engineer Schaffhausen, CH (Remote)

- Contribute to the core development of Lemonate's interactive 3D platform, enabling real-time rendering and deployment of complex browser-based applications.
- Develop core rendering engine features, including real-time particle systems, post-processing pipelines, and low-level graphics architecture.
- Implement advanced rendering techniques and performance optimizations in a WebGL-based engine for cross-platform 3D visualization.
- Drive engine architecture decisions and build tools to streamline 3D workflows across web environments.

Mindcraft AG October 2022 – April 2025
Senior Software Engineer Bern, CH (Remote)

- Designed and implemented a high-performance web rendering engine for the Meshvalley platform.
- Built end-to-end 3D scanning pipelines, including asset processing, optimization, and preparation for real-time rendering.
- Architected systems for hosting, streaming, and interactive visualization of large-scale 3D datasets in the browser.
- Developed geometric algorithms and data-processing workflows to efficiently handle complex scanned environments.

Panua July 2022 – November 2023
Web Developer / DevOps Engineer (Freelance) Lugano, CH (Remote)

- Collaborate with Panua's team on an hourly basis, providing expertise in Web Development, DevOps and Software Engineering.
- Implemented a software to generate, validate, and authenticate users' machine fingerprints to prevent the piracy of Panua Technologies Software, enhancing the security of the company's products.
- Ensuring high availability and performance of Panua's software platforms.
- Develop automated deployment and testing pipelines using Docker, improving the efficiency and reliability of software releases.

Hegias September 2018 – September 2020
Software Engineer | DevOps Manager Lugano & Zurich, CH (Hybrid)

- Led Agile team of 5 developers to continuous deployment and increased productivity by 20% by pair programming.
- Responsible for the Unix server infrastructure capable of handling thousands of clients' request.
- Independently solved the performance issues of the rendering engine by implementing advanced rendering techniques, 3D objects compression and a machine learning 3D object classification.
- Decreased the running time of the 3D objects processing pipeline by more than 250% by distributing the server infrastructure on different nodes.
- The company reached more than 2.1 milion Swiss Francs in funding and was selected as one of the top 10 startup to look after.

Institute of Computational Science June 2018 – September. 2018
Junior Researcher Lugano, CH

- Contributed to the development of Utopia, a state-of-the-art High-Performance C++ parallel computing library to solve non-linear high-dimensional problems.
- Full source code available on the Project Maintainer's repository

EDUCATION

Università della Svizzera Italiana September 2019 – June 2022
Master in Computational Science Lugano, CH

- **Thesis**: Invasive species modeling via relational event models.
- **Coursework**: Stochastic Optimization, Numerical Analysis, PDE/ODE, Deep Learning.
- **GPA**: 7.66

Università della Svizzera Italiana September 2016 – June 2019
Bachelor in Informatics Lugano, CH

- **Project**: A robust pipeline to process 3D objects.
- **Coursework**: Data Structures, Algorithms, Databases, Computer Systems, Machine Learning.
- **GPA**: 7.45

PROJECTS

3D DICOM Visualiser

- A modern, browser-based medical imaging viewer for visualizing DICOM files in 2D and 3D. Free and open-source.
- Built with privacy in mind; all processing happens 100% client-side with no server uploads required.
- Developed as a learning tool to explore DICOM parsing, volumetric rendering, and GPU-accelerated visualization techniques in the browser. Not intended for clinical or diagnostic use.

3D Spectral Segmentation

- Designed and implemented a Blender's add-on to automatically detect 3D objects features using a custom Spectral Segmentation algorithm.
- The add-on received more than 200 downloads alongside a lot of positive feedback.
- Full source code can be found on my personal GitHub profile.

SKILLS

Programming languages:
Python C++, C#, OO
Javascript, Rust

Frameworks:
Numpy, Pandas, Scikit-learn, PyTorch, Cuda, OpenMPI, NodeJS, mongoDB, WebGL, ThreeJS

Deployments:
AWS, Kubernetes, Rancher, Docker, Linux

Languages:
Italian: Native
English: Full Professional Proficiency
Spanish: Conversational Proficiency
German: Basic Proficiency